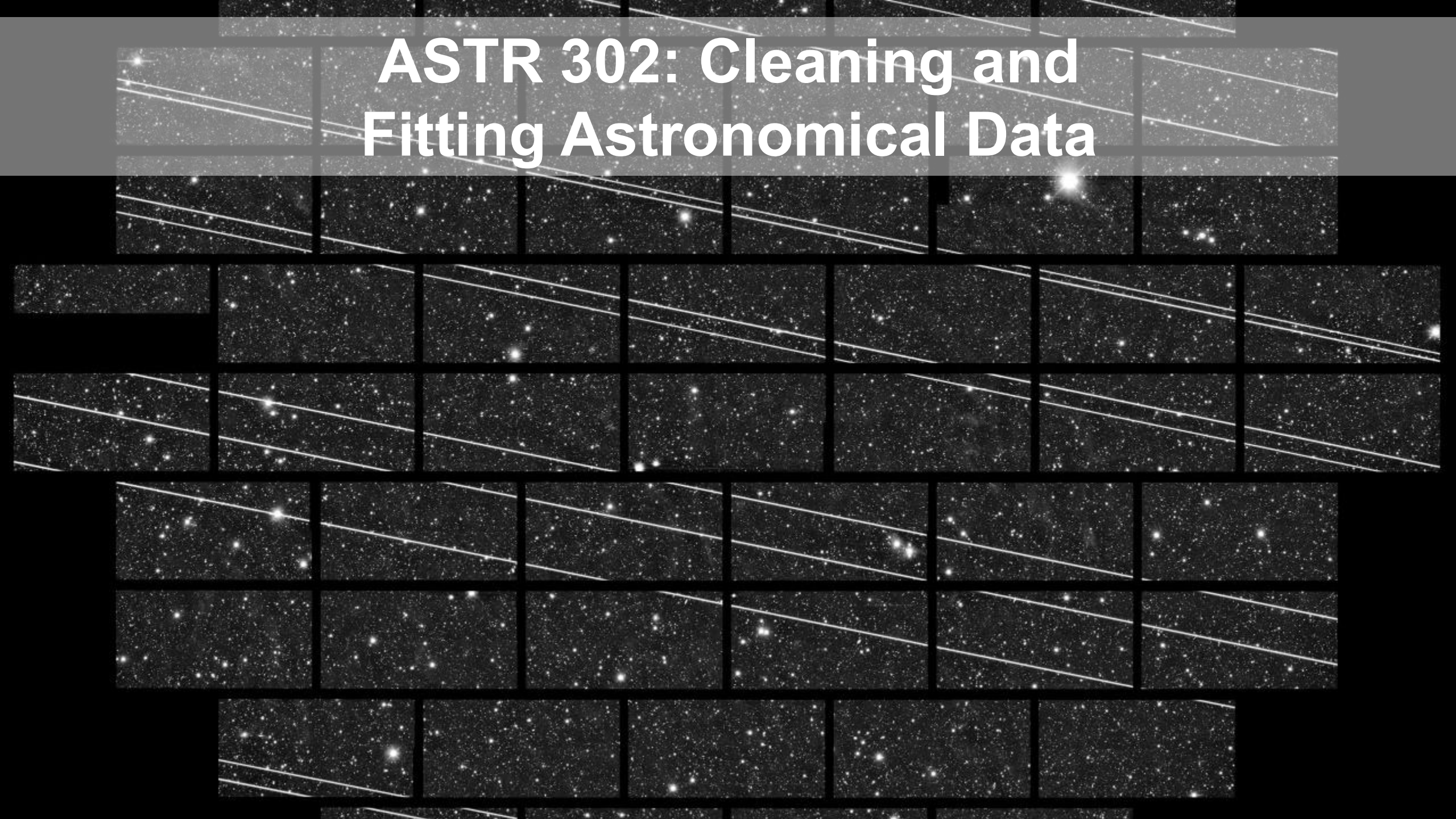
ASTR 302: Cleaning and Fitting Astronomical Data



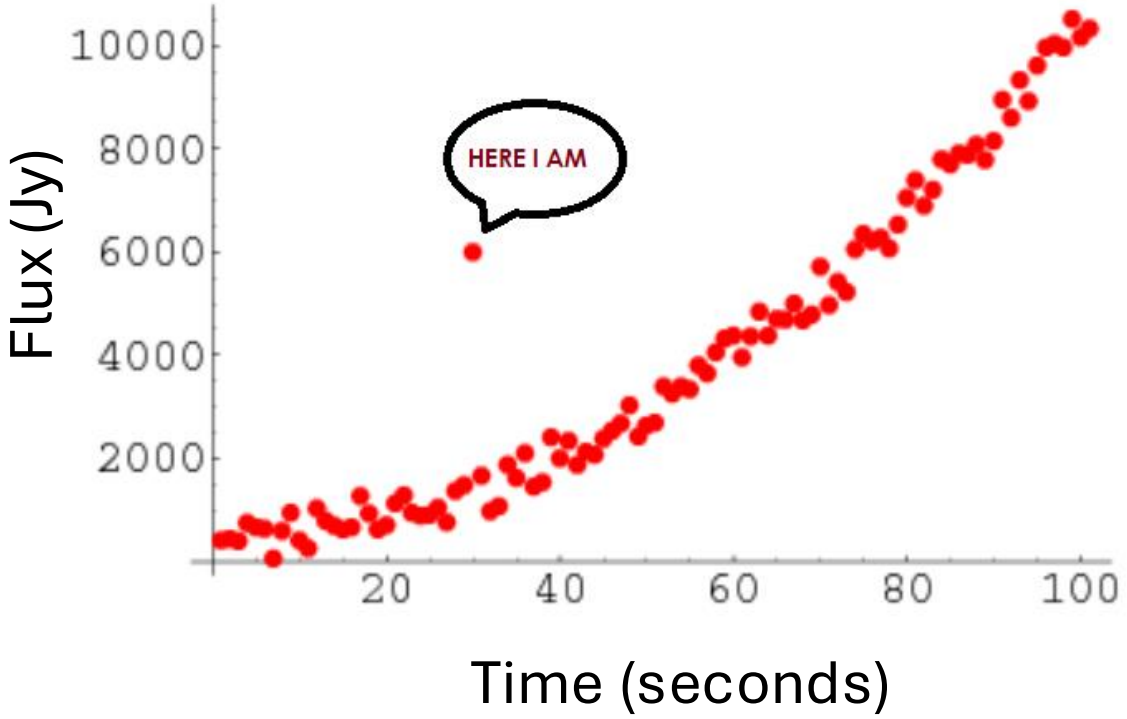
Agenda – November 9th , 2026

- NIMOY updates (we are moving to NIMOY2)
- Rejecting Bad Data
- Fitting Astronomical Data

Relevant Reading: Chapter 6, 7, 8 (~20 pages total)

Summary of Scientific Paper Due: November 13th, 2026
Submit Initial Result Plot: November 20th , 2026
YOU NEED TO BE HERE: November 20th, 2026

Outliers in Measurements



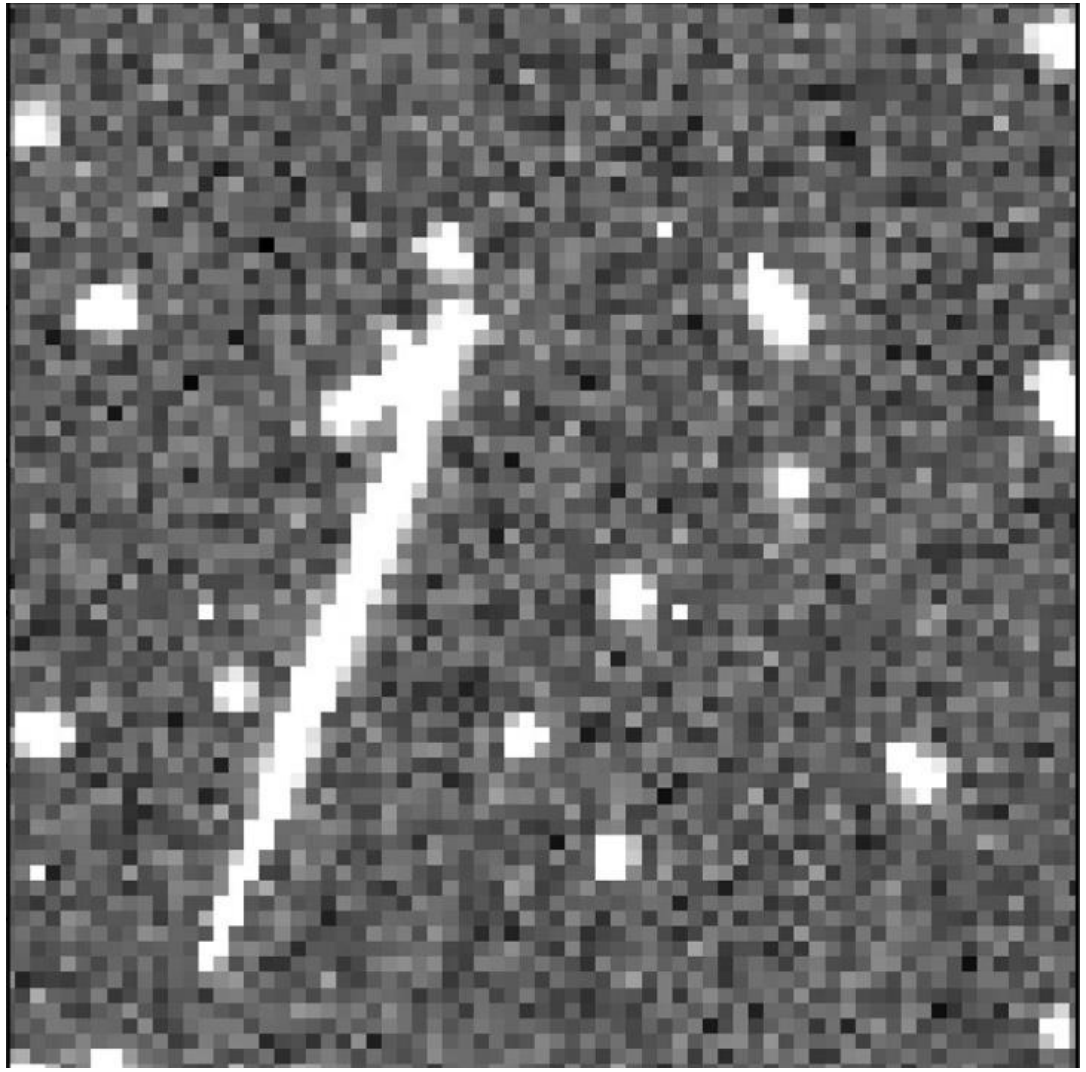
Example: Outlier in 1-D data

Outliers are measurements or data points that deviate significantly from some mean or expected behavior.

It can be challenging to identify outliers, but removing them can often improve the precision and/or accuracy of your measurements (**Lecture 2**).

Depending on the sample size (N) of your data there are several techniques for removing outliers.

Outliers in Measurements



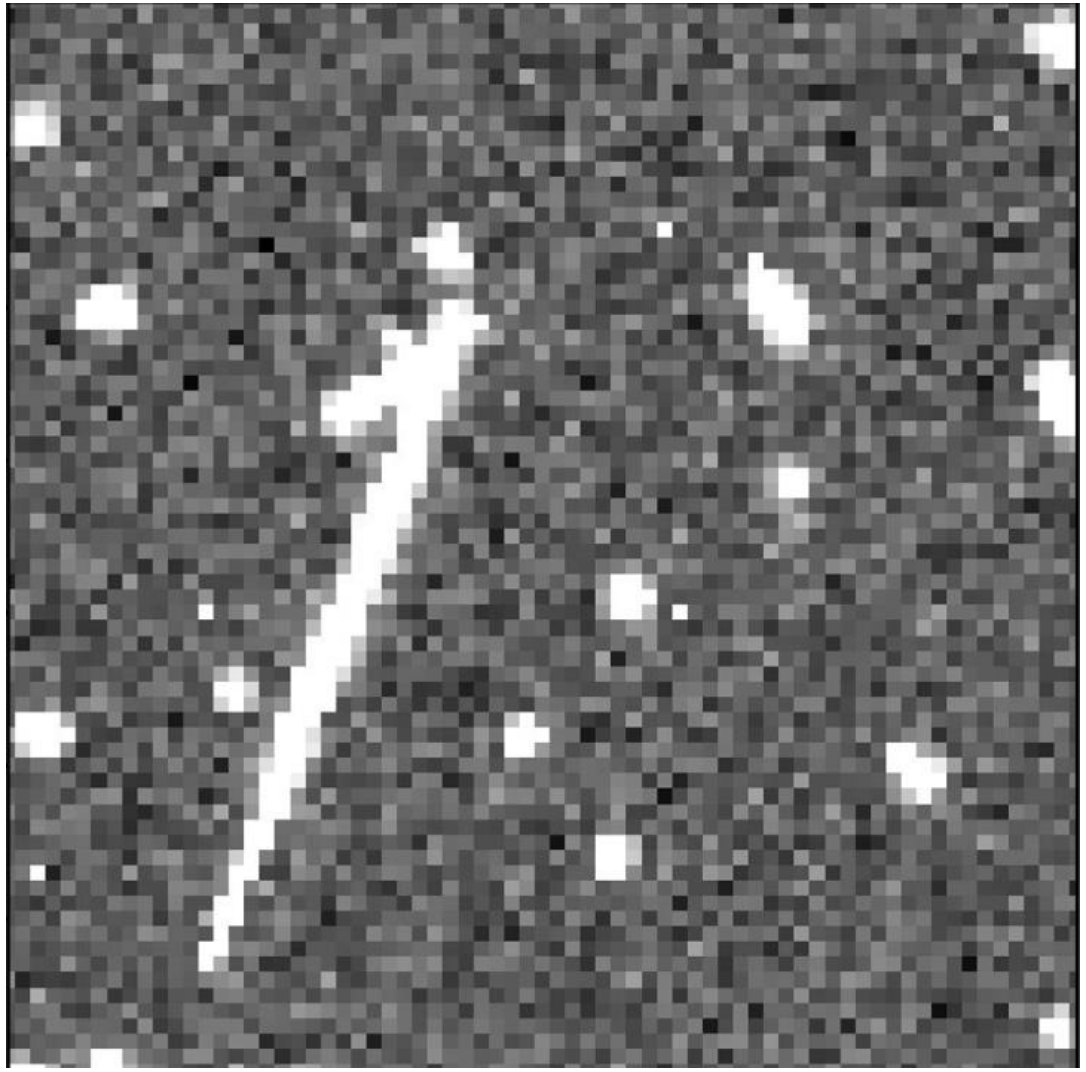
Example: Outliers in 2-D data

Outliers are measurements or data points that deviate significantly from some mean or expected behavior.

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Outliers in Measurements



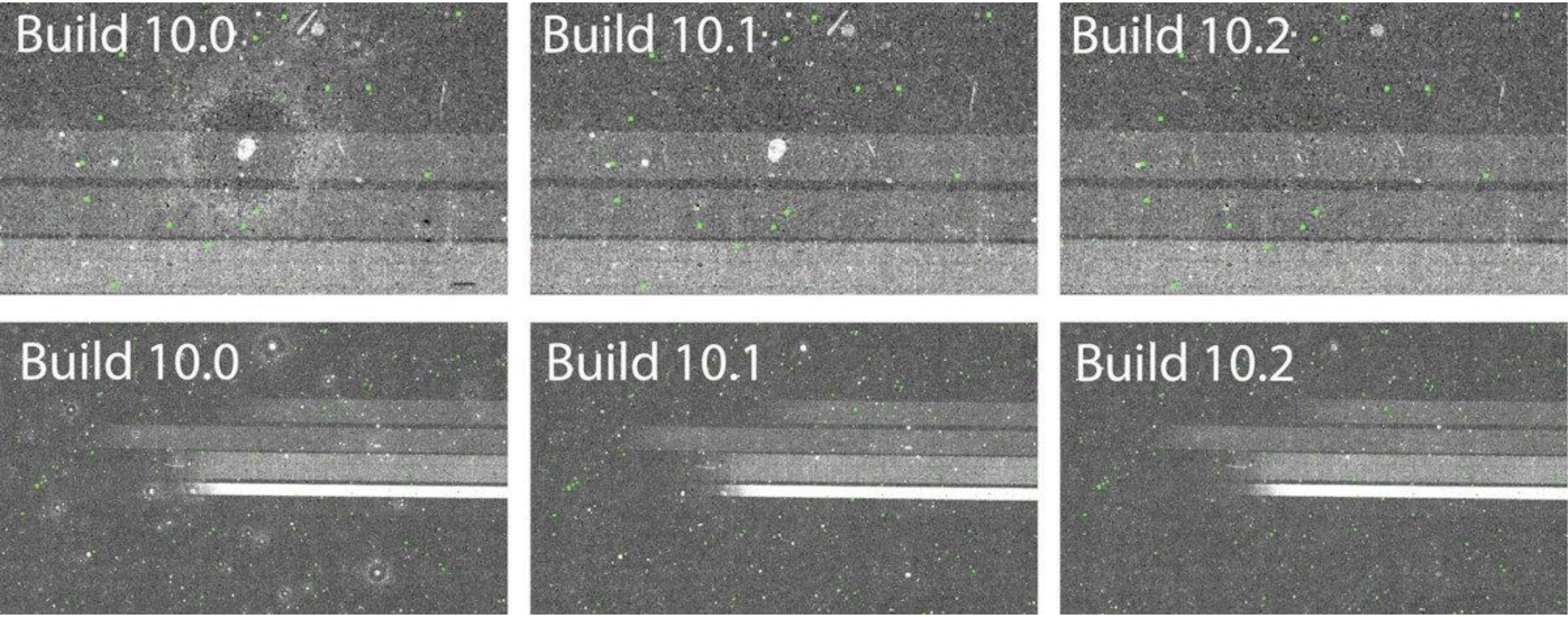
HST/ACS Cosmic Rays
https://www.stsci.edu/hst/instrumentation/acs/performance/anomalies-and-artifacts

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Outliers in Measurements



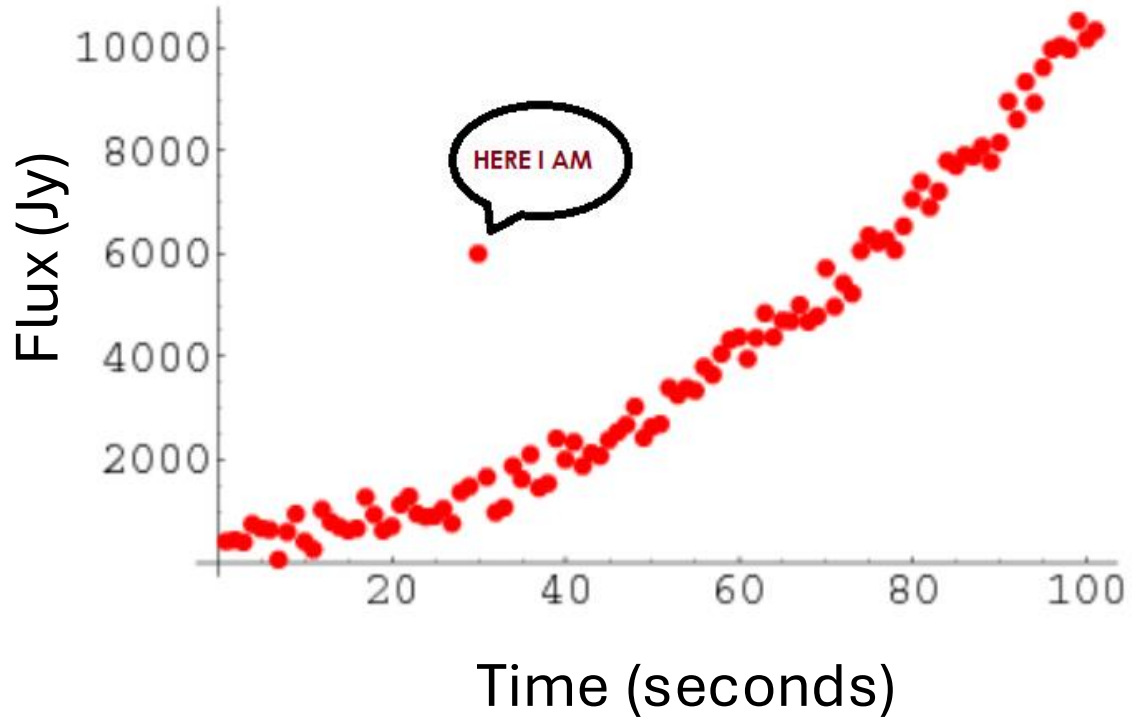
Example: Outliers in 2-D data

JWST/NIRSpec Cosmic Ray Snowball Artifacts

https://www.stsci.edu/contents/news/jwst/2024/jwst-pipeline-news-updated-corrections-for-cosmic-ray-snowball-artifacts

Removing Outliers in Measurements

Removing outliers is not an exact science, but you need to try your best to do it as uniformly as possible.



Example: Outlier in 1-D data

Removing Outliers in Measurements

Removing outliers is not an exact science, but you need to try your best to do it as uniformly as possible.

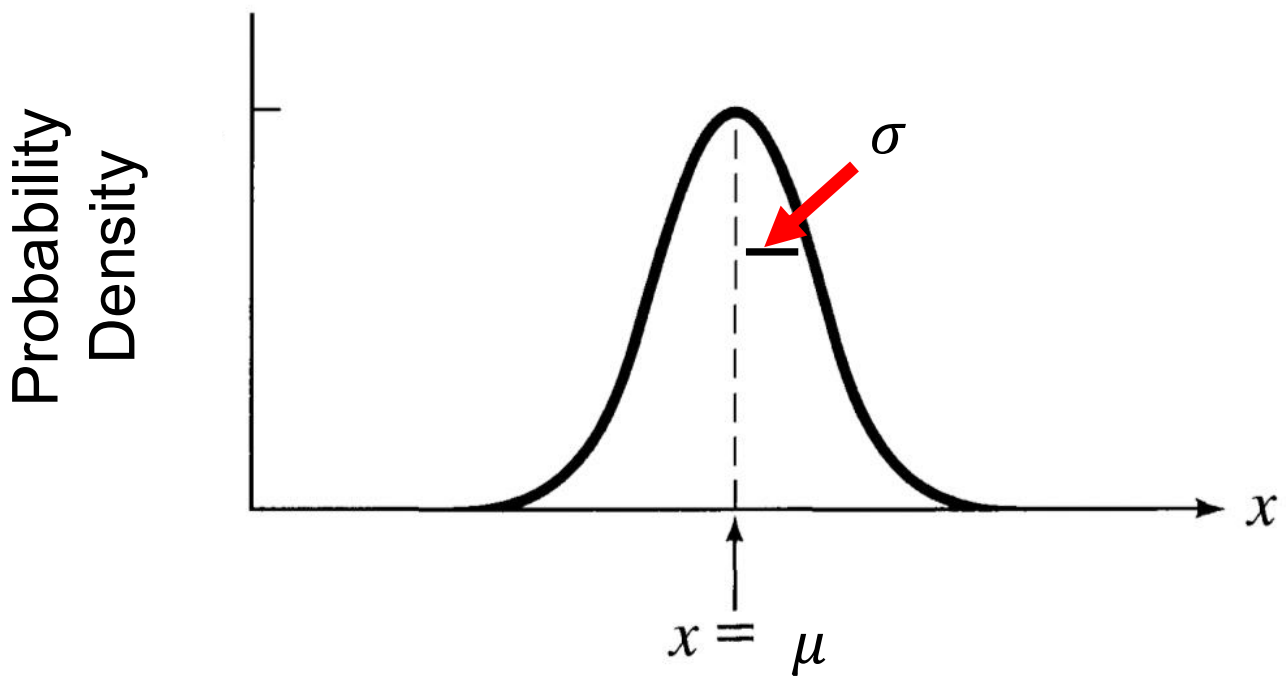
There are several algorithms to remove outliers but we will discuss two*.

-Chauvenet's criterion (**Chp 6.**)

-Sigma clipping

***These methods assume your measurements follow Gaussian/Normal distribution behavior**

Review: The Normal Distribution

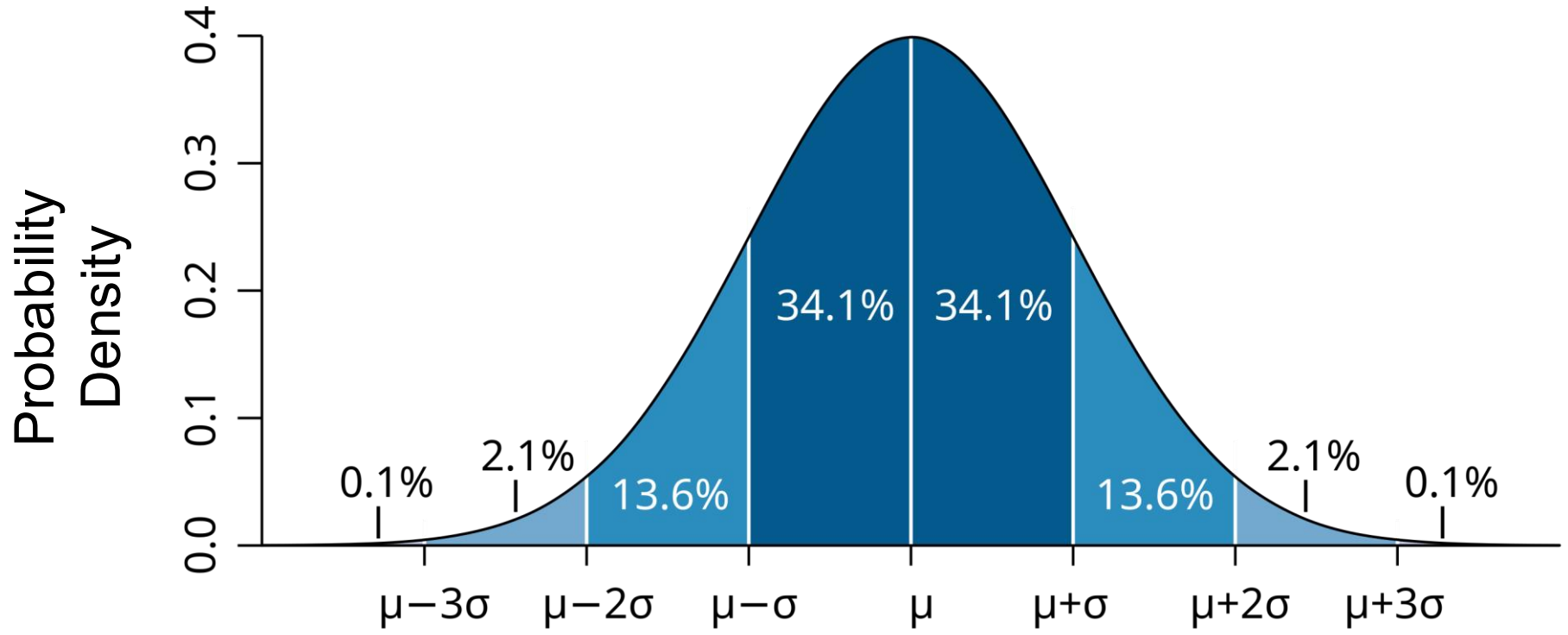


The Normal distribution or the Gaussian distribution is a bell-shaped probability curve centered on the value μ with a width parameter σ

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

μ – expected value
σ – standard deviation

The Normal Distribution: Confidence Intervals



Between $\boldsymbol{\mu - \sigma}$ and $\boldsymbol{\mu + \sigma}$, 68.2% of the data is contained within this interval

Between $\boldsymbol{\mu - 2\sigma}$ and $\boldsymbol{\mu + 2\sigma}$, 95.4% of the data is contained within this interval

Between $\boldsymbol{\mu - 3\sigma}$ and $\boldsymbol{\mu + 3\sigma}$, 99.7% of the data is contained within this interval

Chauvenet's Criterion for Outlier Rejection

- Calculate the mean **(μ)** of your measurements **(x)**
- Calculate the standard deviation **(σ)** of your measurements
- Take the absolute difference between the measurements and the mean **($\bar{x}$)**

$$|x - \bar{x}|$$

- Take the ratio of the absolute difference to the standard deviation

$$\textbf{Ratio} = \frac{|x - \bar{x}|}{\sigma}$$

- The "suspect" point or the likely outlier has the biggest ratio value (t_{sus} or R)

$$\textbf{Ratio}_{\textbf{max}} = t_{sus} = \frac{|x_{sus} - \bar{x}|}{\sigma}$$

See Chauvenet_criterion.ipynb on class github

Chauvenet's Criterion for Outlier Rejection

- Estimate the probability of a measurement residing inside $(t_{sus} * \sigma)$

$$\textbf{P(inside)} = \int_{-t_{sus}\sigma}^{t_{sus}\ \sigma} P(x)$$

- Calculate the probability the point is outside of this range

P(outside) = 1 – P(inside)

- Calculate expected deviation **(n)**

$$n = N_{measurements} * \textbf{P(outside)}$$

If n < 0.5 (low chance of happening) remove the suspect data point.

See Chauvenet_criterion.ipynb on class github

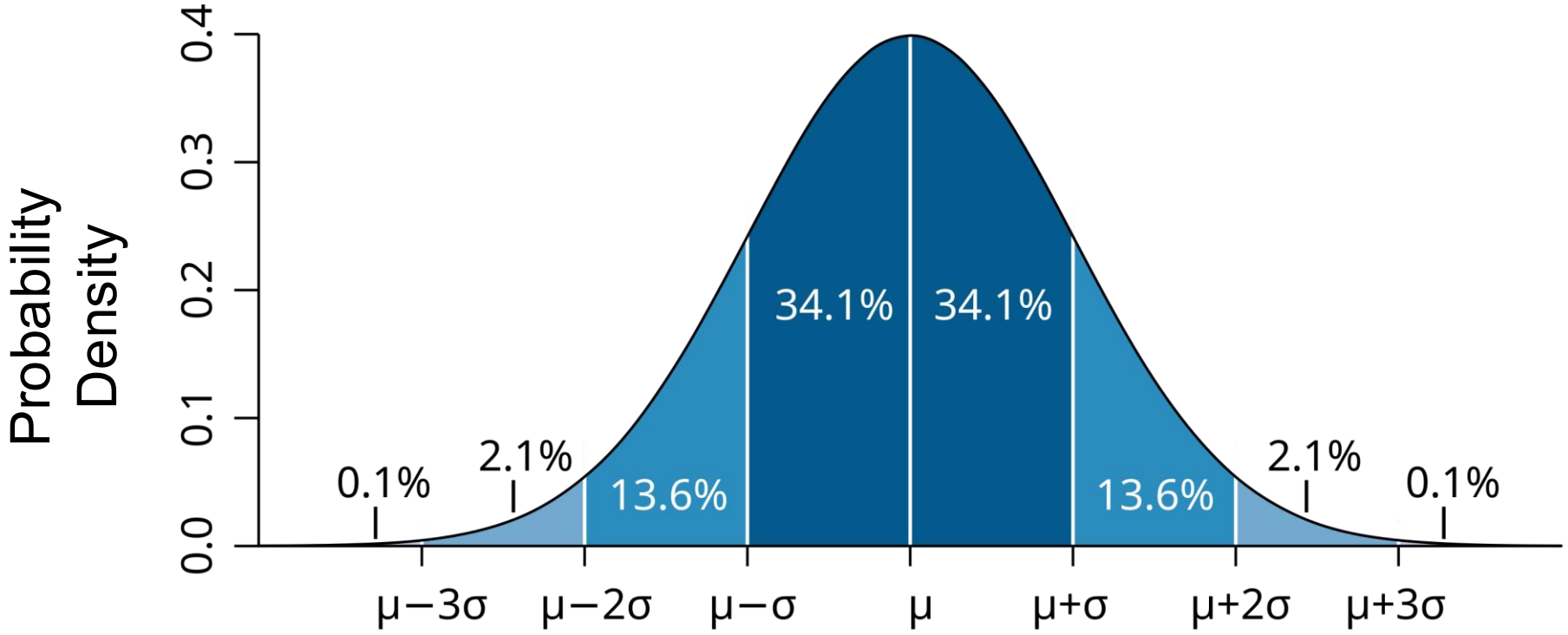
Chauvenet's Criterion: Caveats

- Not really great for small sample sizes, the outlier is already included in the mean and original sigma

- The 0.5 threshold is arbitrary

See Chauvenet_criterion.ipynb on class github

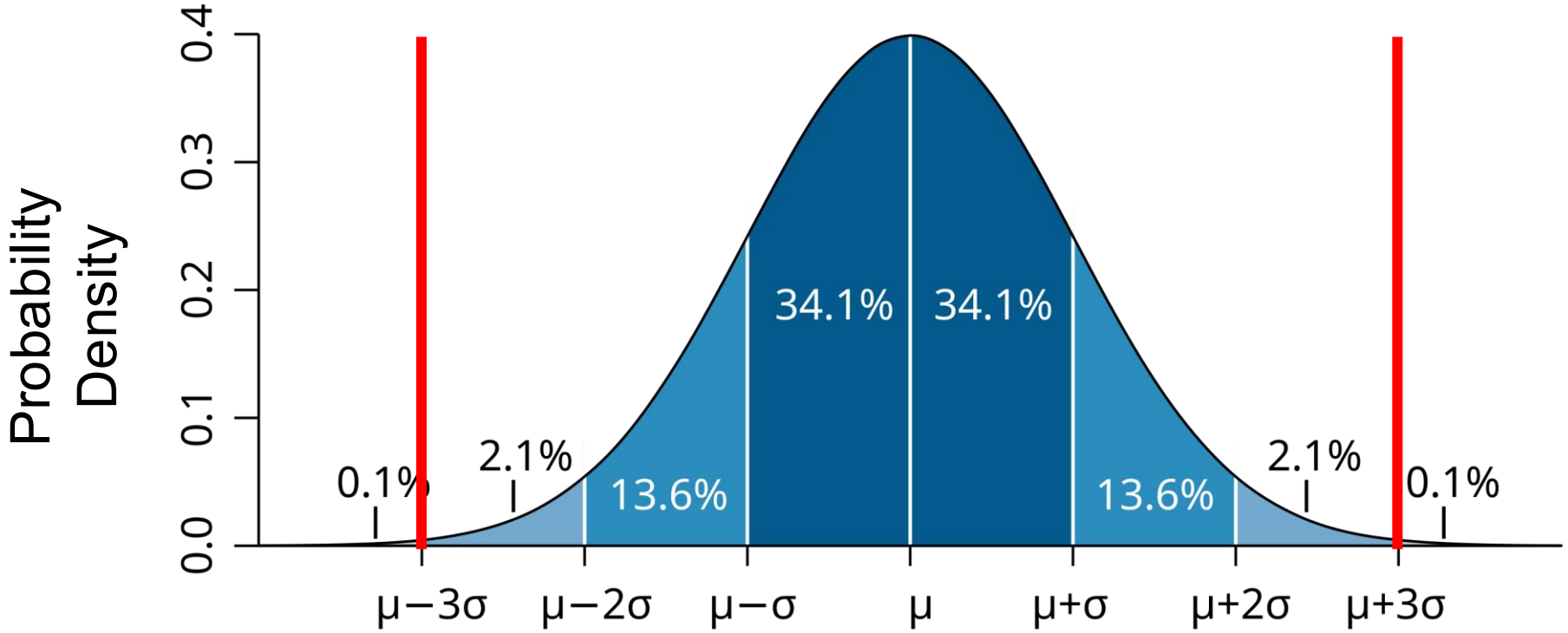
Sigma Clipping



- You can choose some sigma threshold ahead of time and just clip points outside of that bounds.

- Scipy and Astropy have functions to do this once and iteratively.

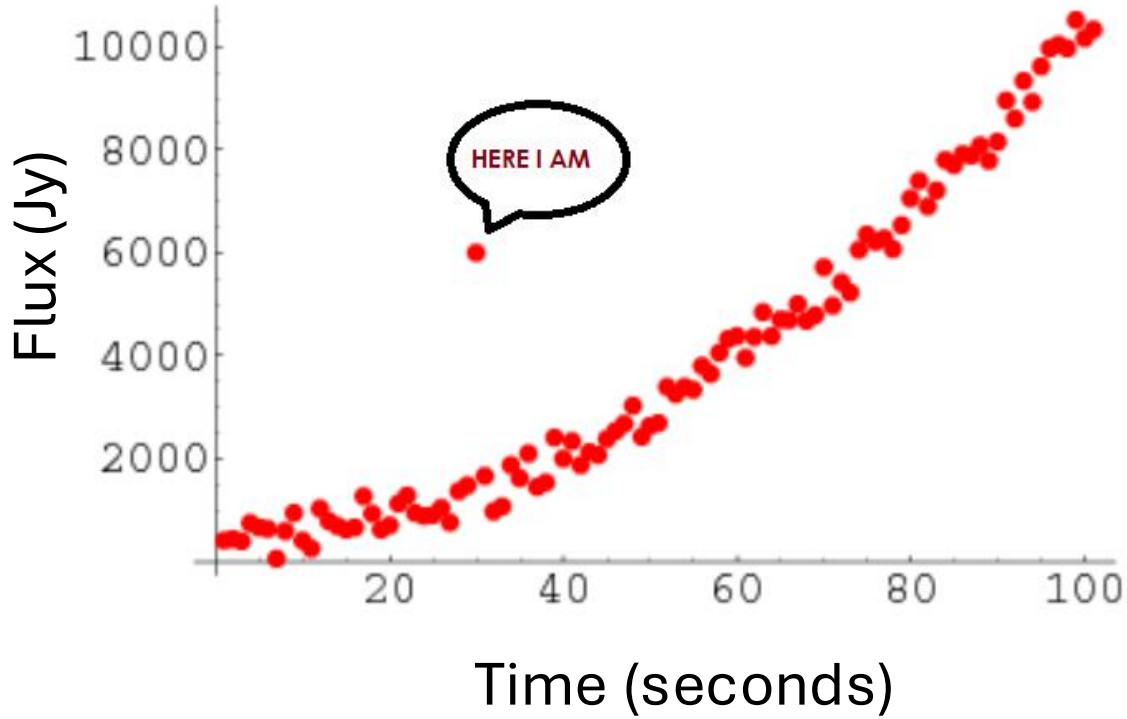
Sigma Clipping



- You can choose some sigma threshold ahead of time and just clip points outside of that bounds.

- Scipy and Astropy have functions to do this once and iteratively.

Sigma Clipping: Caveats



- Often you need to do sigma clipping within a given region/area
- With any clipping of data you should probably look at the uncleaned version first

Chi Squared Test

$$\chi^2 = \frac{(measurement - expectation)^2}{\sigma^2}$$

Chi Squared Test is a way to assess the goodness of your data to some expectation, model, or template.

Chi Squared Test

$$\chi^2 = \sum_{i=0}^{N} \frac{(measurement_i - expectation_i)^2}{{\sigma_i}^2}$$

Chi Squared Test is a way to assess the goodness of your data to some expectation, model, or template. The test can be completed over several data points.

Least Squares Fitting

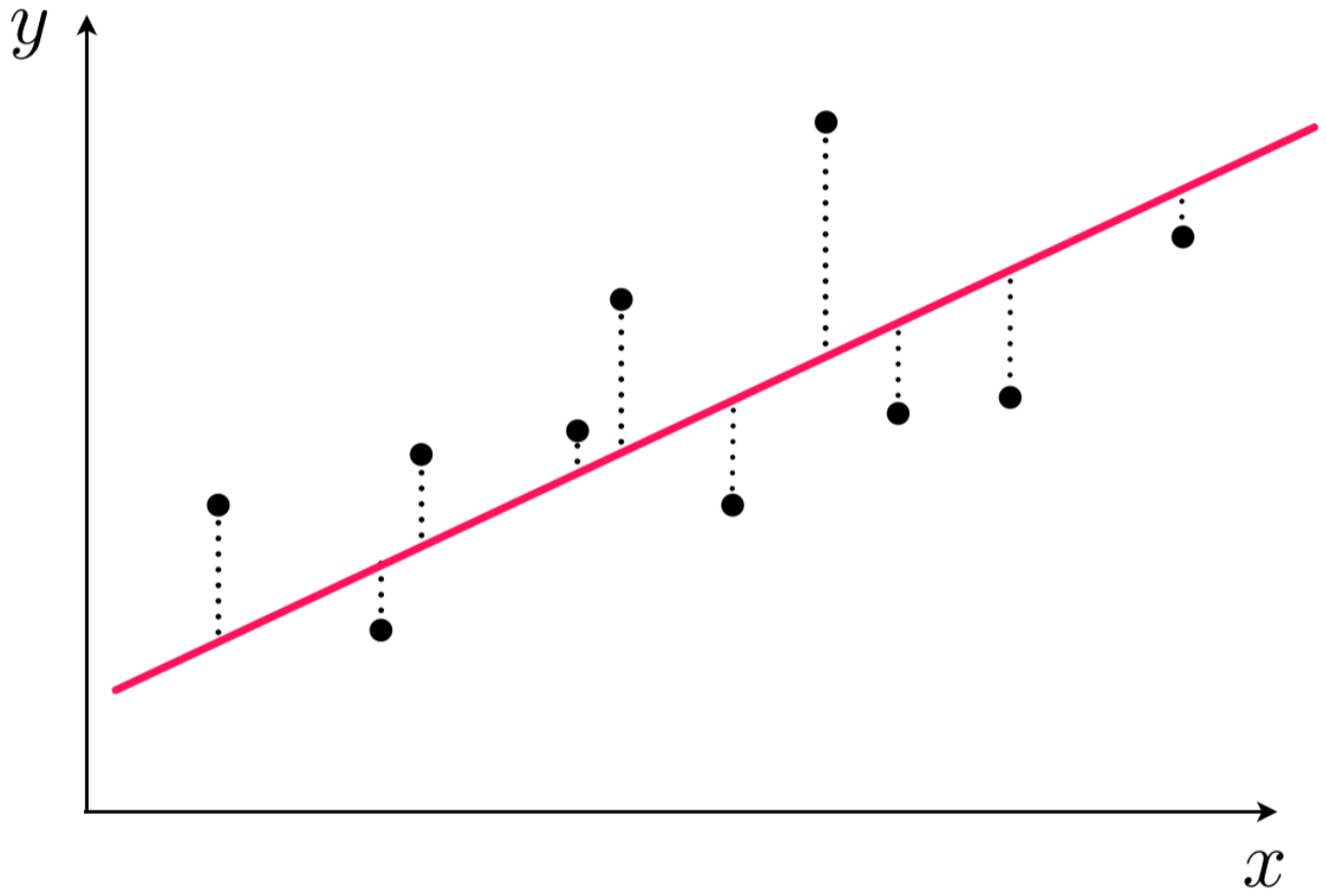
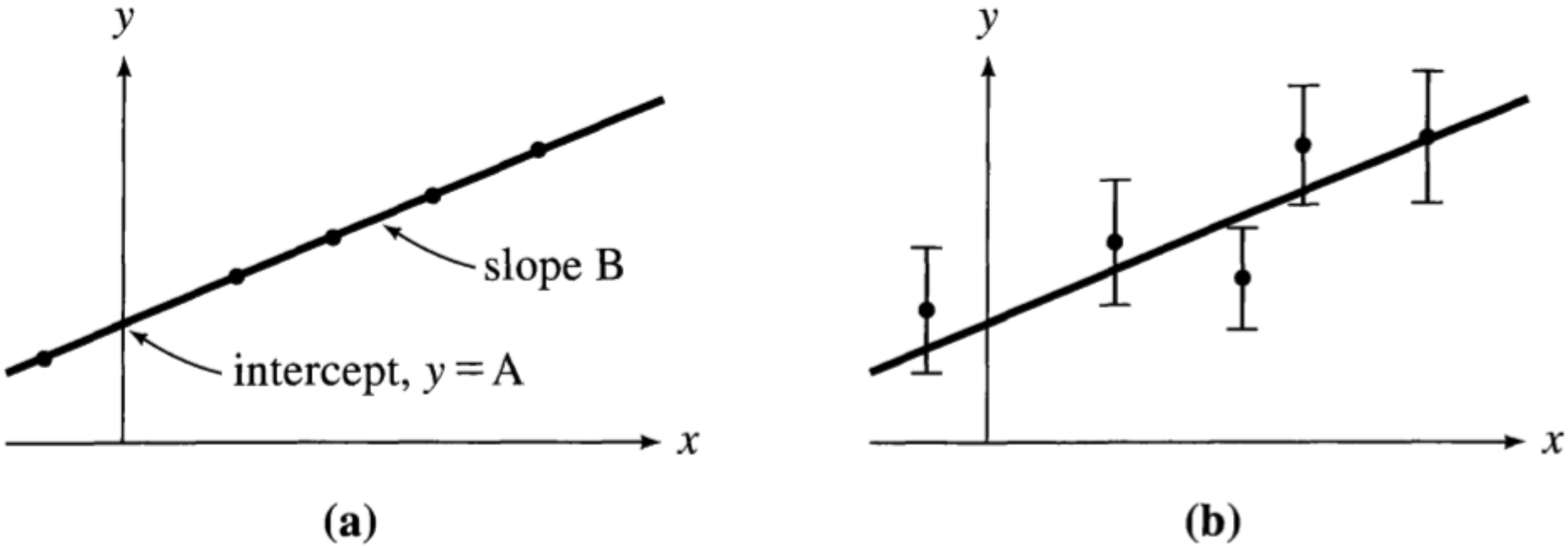


Figure from
https://kenndanielso.github.io

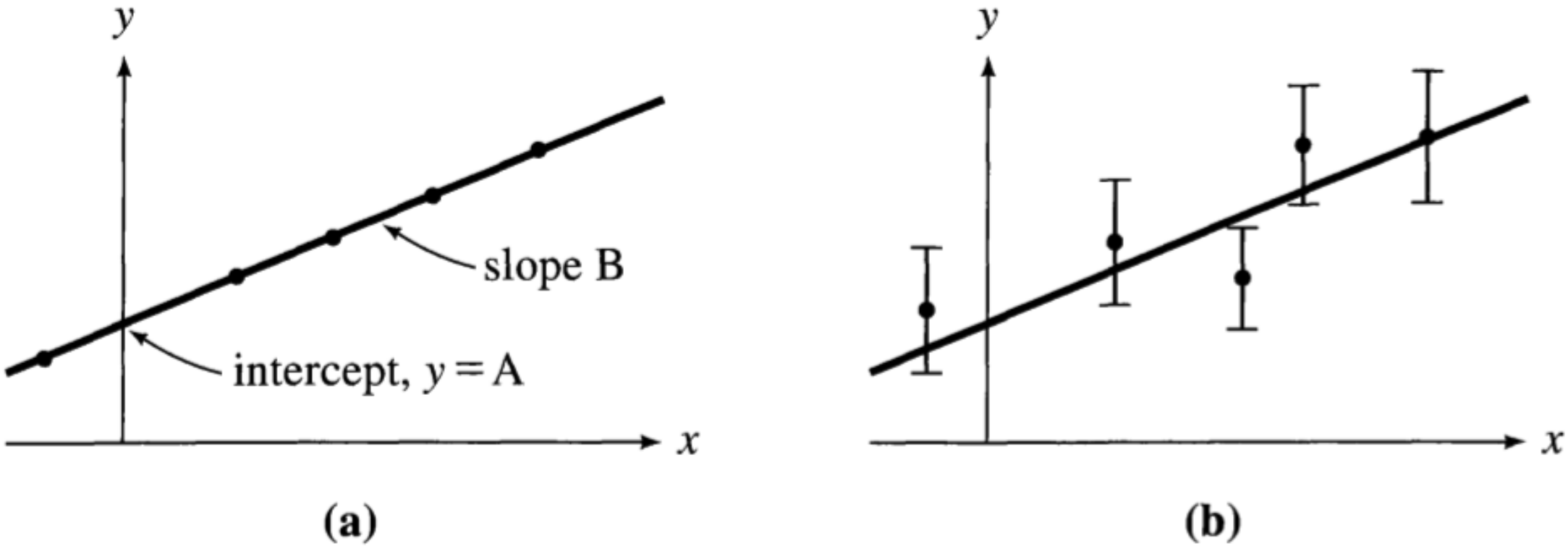
Least squares fitting is a linear regression technique to find a relationship between two or more variables. The goal is to find a line or solution that minimizes the residual between your data and the expected value.

Least Squares Fitting



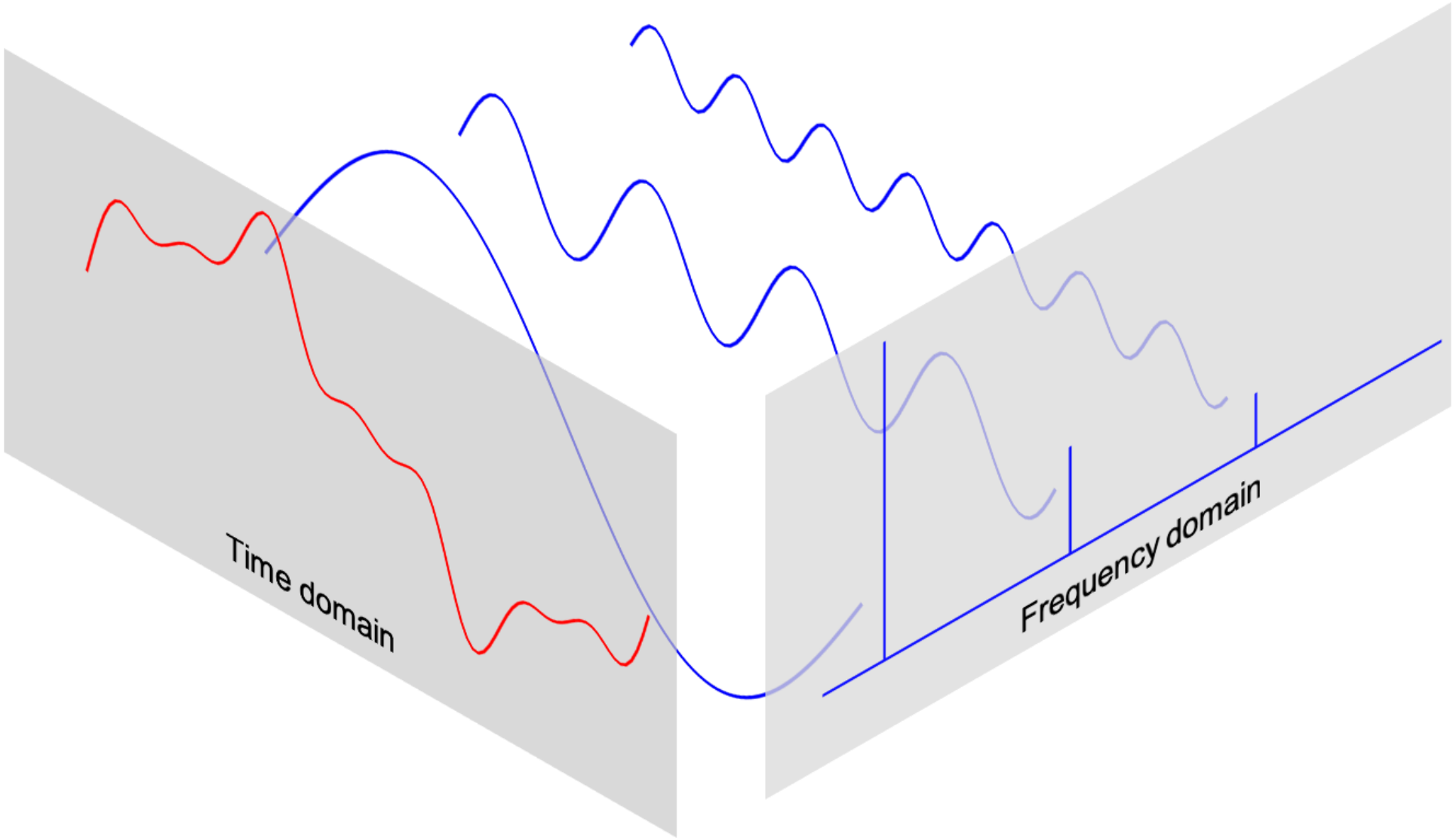
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Least Squares Fitting: Line Example

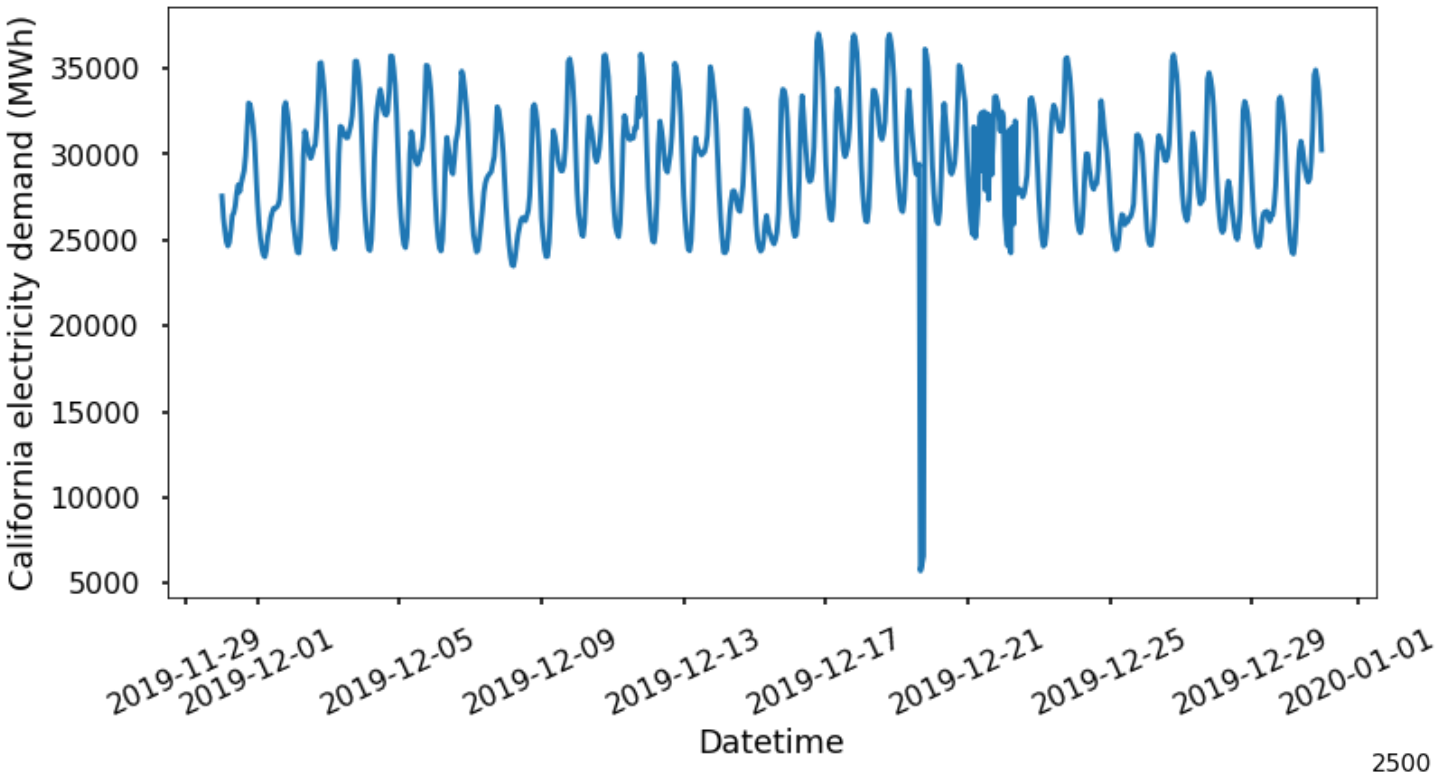


If we assume our data points follow Gaussian statistics, we can solve for **A** and **B** and estimate their errors.
(Taylor, Chapter 8)

Estimating Periodicity in Time-Series Data



Estimating Periodicity in Time-Series Data

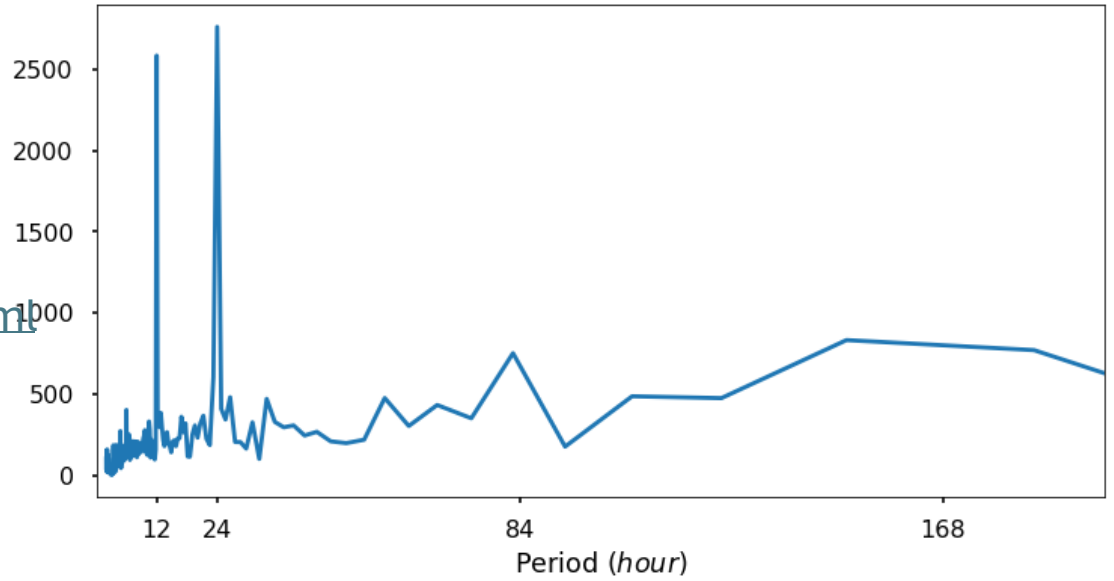


Fast Fourier Transform

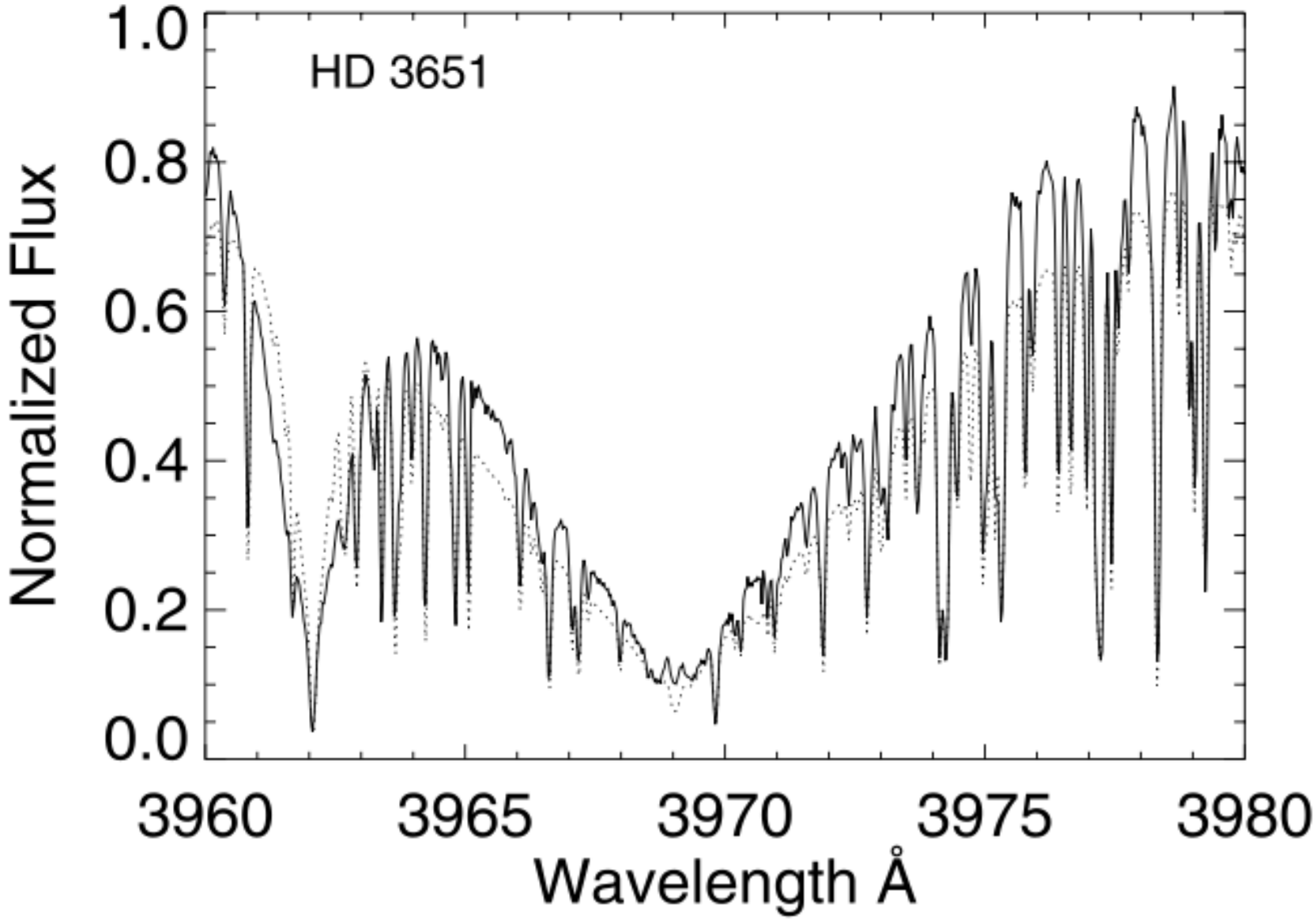
https://zingale.github.io/computational_astrophysics/ffts/dft.html

scipy.fftpack import **fft, fftfreq**

Evenly spaced data

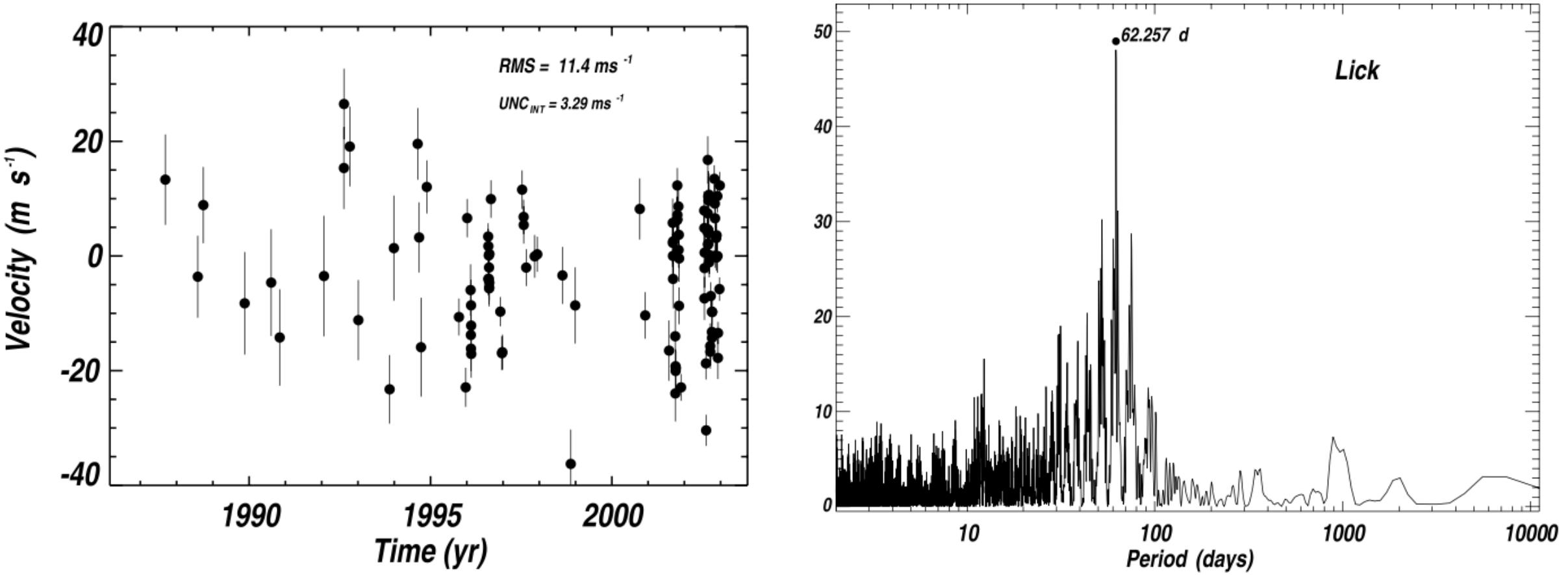


Estimating Periodicity in Time-Series Data



Fischer+ 2003

Estimating Periodicity in Time-Series Data

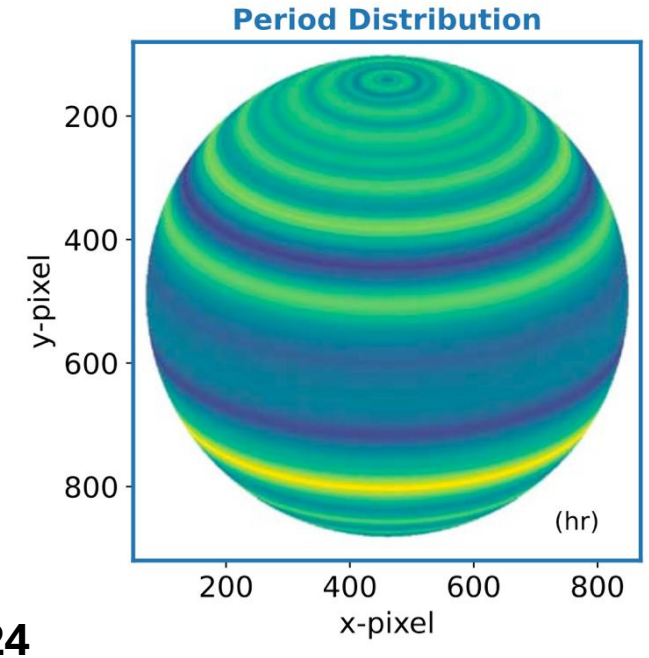


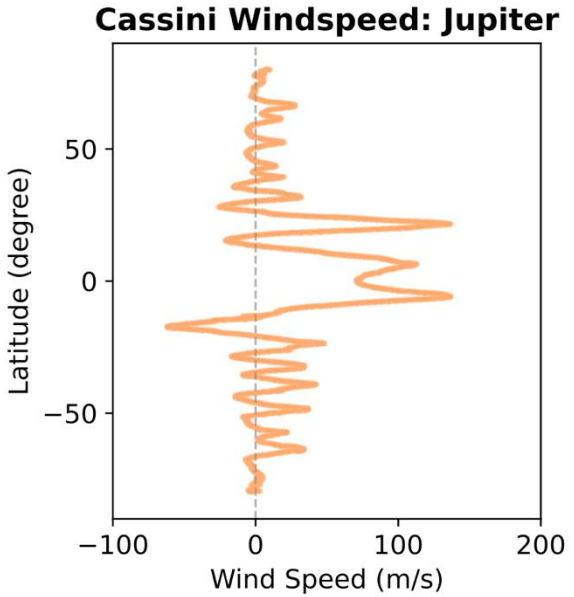
LombScargle Periodogram
from **astropy.timeseries** import **LombScargle**
Even and unevenly spaced data

Fischer+ 2003

See VanderPlas + 2022

Estimating Periodicity in Time-Series Data

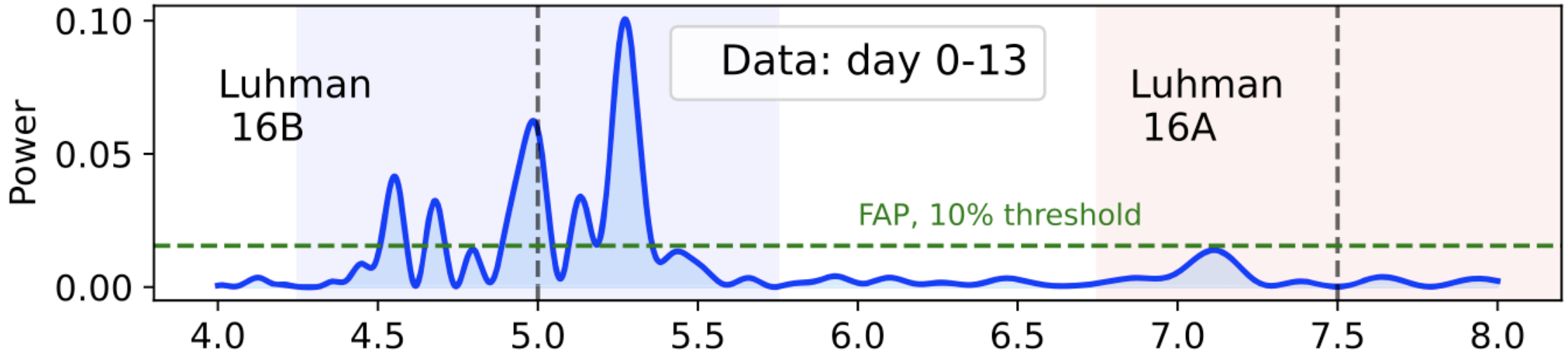




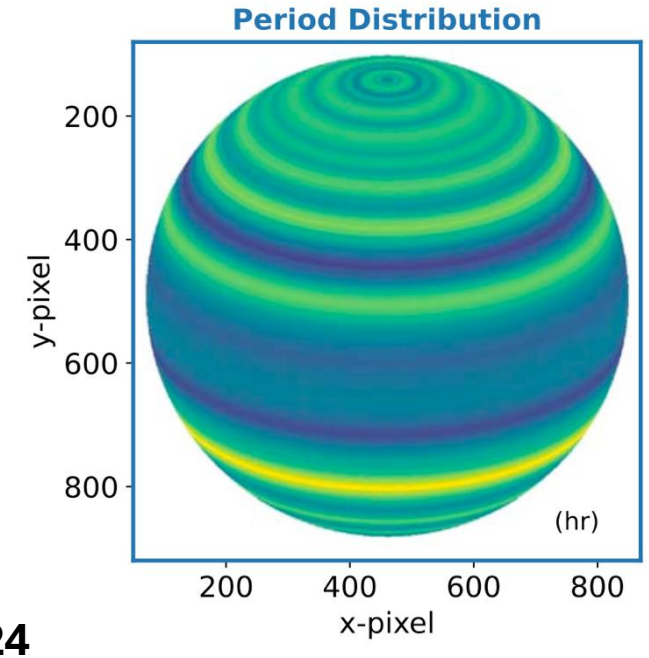
Jupiter and Brown Dwarfs have winds that create a variety of observed periods in time series photometry.

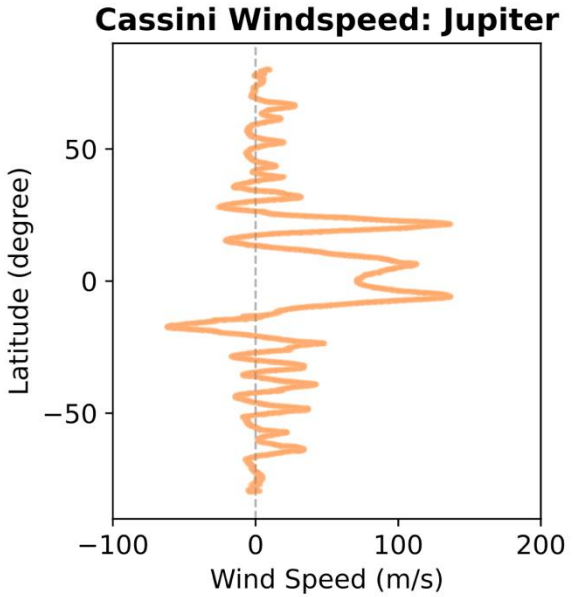
LS = LombScargle(time, flux, flux_err)
LS.false_alarm_probability()

Fuda+ 2024



Estimating Periodicity in Time-Series Data





Jupiter and Brown Dwarfs have winds that create a variety of observed periods in time series photometry.

Fuda+ 2024

